AI-BASED ELECTRICITY DEMAND INTELLIGENCE SYSTEM FOR SRI LANKA

INDIVIDUAL THESIS

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Individual component: Explainable AI for Electricity Demand Forecasting

Group ID: R26-IT010

Report submitted in partial fulfilment of the requirements for the

Bachelor of Science (Hons) in Information Technology

Specialising in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Accurate electricity demand forecasts are essential for power system planning, yet the machine-learning models that produce them are usually black boxes: they return a number without saying why. This thesis presents the Explainable AI (XAI) component of the AI-Based Electricity Demand Intelligence System for Sri Lanka (Project R26-IT010). The component makes the predictions of an electricity demand forecasting model understandable by combining two explanation methods, SHAP for global explanations and LIME for local explanations, and by delivering them through a FastAPI backend and an interactive dashboard that is linked to anomaly detection.

The study uses 189,888 electricity demand observations recorded at 15-minute intervals from January 2020 to May 2025, together with weather, economic and time features. After target-only IQR outlier removal and a chronological 80/20 split, Linear Regression, Random Forest and XGBoost were compared. Linear Regression and Random Forest were more accurate on this near-linear dataset, but XGBoost was selected for explanation because SHAP's TreeExplainer gives exact values for tree-based models. XGBoost was tuned with Optuna (50 trials, TPE sampler) without using the test set and reached RMSE 2.02 kW, MAE 1.05 kW and R^2 0.9999 on the test set, with training and test errors almost identical.

SHAP was computed separately for each year from 2020 to 2025. Temperature was by far the most influential feature in every year (mean absolute SHAP about 160 kW in 2020 and about 157 kW in 2025, against about 0.2 to 0.3 kW for the other features), and drift between years was small. LIME explained 15 instances sampled from low, medium, high and peak demand bands; the local R^2 was about 0.60 for the low, high and peak instances but only about 0.04 for the four medium-demand instances. The explanations are served by two API endpoints to two dashboard pages and connected to the behaviour intelligence dashboard, so that a flagged anomalous day opens its explanation. Limitations include LIME coverage of only 15 instances, a near-linear dataset that appears to be synthetic, low LIME fidelity for medium-demand instances, and the absence of a user evaluation.

Keywords: Explainable AI, SHAP, LIME, Electricity demand forecasting, XGBoost, Temporal drift

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my project supervisor, Dr. Nathali Silva, and my co-supervisor, for their valuable guidance, continuous support and constructive feedback throughout this research. Their encouragement greatly helped me to understand the research problem and improve the quality of my work.

I am also thankful to the Department of Information Technology and the Sri Lanka Institute of Information Technology (SLIIT) for providing the academic environment, learning resources and technical support needed to carry out this research.

My heartfelt appreciation goes to my project team members, Thanujan T (IT22545176), Janahan R (IT22069740) and Dewmini D.M.D.P (IT22630216), for their cooperation, dedication and teamwork. Their collaboration, especially the integration between the Behaviour Intelligence dashboard and my research component, played an important role in the progress of this research.

Finally, I would like to thank my family and friends for their continuous encouragement, understanding and moral support throughout the completion of this research.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
B2G	Business-to-Government
CEB	Ceylon Electricity Board
GDP	Gross Domestic Product
GRU	Gated Recurrent Unit
IQR	Interquartile Range
JSON	JavaScript Object Notation
LECO	Lanka Electricity Company
LIME	Local Interpretable Model-agnostic Explanations
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
ML	Machine Learning
R^2	Coefficient of Determination
REST	Representational State Transfer
RMSE	Root Mean Squared Error
RQ	Research Question
SDG	Sustainable Development Goal
SHAP	SHapley Additive exPlanations
SLIIT	Sri Lanka Institute of Information Technology
TPE	Tree-structured Parzen Estimator
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting

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Appendix	Description
Appendix A	Extended Results and Evidence Record
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1 INTRODUCTION

1.1 Overall Project Context

Electricity cannot be stored cheaply at grid scale, so generation has to follow demand almost instantly. Demand in Sri Lanka changes with the hour of the day, the day of the week, the season, public holidays and weather, and it is also shaped by economic conditions such as income growth and the price of electricity [1], [3]. Poorly anticipated changes lead to excess or insufficient supply, higher operating costs and power outages. Demand forecasting has therefore become an integral part of modern energy management, and artificial intelligence (AI) is now widely used to forecast demand more accurately [1], [2].

The research project **AI-Based Electricity Demand Intelligence System for Sri Lanka** (Project ID R26-IT010) responds to this need. Rather than offering one prediction model on its own, the project aims to provide a decision-support system in which four connected services work on the same data: behaviour analysis of consumption, adaptive demand forecasting, explanation of model predictions, and scenario-based decision support. The system is implemented in Python, and its services are exposed through a FastAPI backend to a web dashboard for energy planners and analysts. All four services use the same underlying dataset of electricity demand, weather and economic information recorded every 15 minutes from January 2020 to May 2025.

The work is carried out by a team of four undergraduates, each responsible for one component. Table 1.1 summarises the components and shows where this thesis fits. The overall architecture adopted by the team is shown in Figure 1.1.

Table 1.1: Components of the complete solution and the component reported in this thesis

Component (service)	Main purpose	Responsibility
Behaviour intelligence service	Clustering of daily consumption patterns and detection of abnormal demand days	Thanujan T (IT22545176)
Adaptive forecasting and prediction service	Short-term demand forecasting and prediction of future load	Janahan R (IT22069740)
Explainable AI (XAI) service	**Explanation of demand predictions with SHAP and LIME, temporal analysis of feature influence and anomaly diagnosis**	**Madushani R.P (IT22273062) (this thesis)**

Component (service)	Main purpose	Responsibility
Scenario-aware decision support service	Evaluation of demand under selected weather and event scenarios	Dewmini D.M.D.P (IT22630216)

Source: Project proposal, R26-IT010 (Section 4.4).

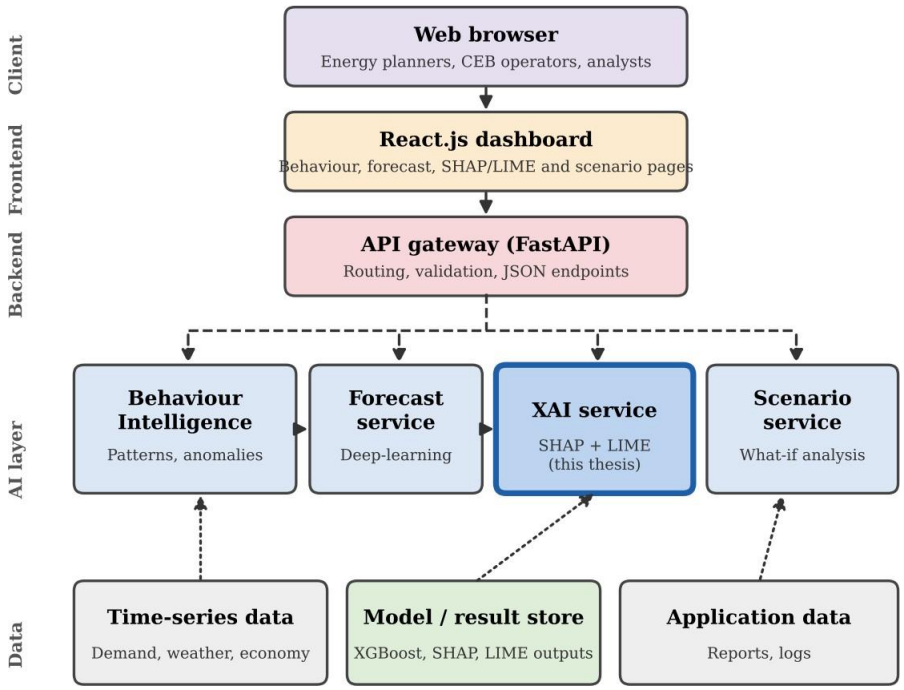


Figure 1.1: High-level architecture of the AI-Based Electricity Demand Intelligence System

Source: Redrawn for this thesis from the project design. The XAI service, outlined in blue, is the subject of this thesis.

1.2 Individual Component Context

This thesis reports the **Explainable AI (XAI) component** of the system. Its task is not to forecast demand better than any other model. Instead, it takes a demand forecasting model and answers a different question: *why did the model predict this number?* It does this with two well-known explanation techniques, SHAP (SHapley Additive exPlanations) [10] and LIME (Local Interpretable Model-agnostic Explanations) [11].

The two techniques are used for different purposes. SHAP is applied at the level of the whole model, separately for each year from 2020 to 2025, to show which input features drive demand and how their influence changes over time (temporal drift). LIME is applied at the level of a single prediction, to explain why the model produced a particular value for a particular 15-minute reading. Fifteen readings were sampled at random from four demand bands (3 low, 4 medium, 4 high and 4 peak) so that the explanations cover a wide range of situations.

The component is connected to the behaviour intelligence component. When the behaviour dashboard flags a day with unusual electricity demand, it offers a link that opens the LIME Anomaly Diagnosis page of this component for that date, so that a planner can move from *an unusual day was detected* to *this is what the model was responding to* in one step. The explanation results are served by a small FastAPI backend and shown on two dashboard pages, SHAP Temporal Drift and LIME Anomaly Diagnosis.

1.3 Problem Statement

A machine-learning model on its own is often called a black box: it receives numbers and returns a prediction but does not say why. For electricity planning this is a real problem. If a model predicts a sudden spike in demand, an energy planner needs to know whether it is caused by hot weather, a public holiday or something else before trusting the prediction enough to act on it [4], [5]. Three weaknesses limit the usefulness of forecasting models in this setting:

- **Opaque predictions.** Accurate models report a demand value without giving the reasons, which reduces the confidence of planners who must justify decisions during peak demand or unusual events.
- **Explanations that are one-off and single-method.** Many studies apply one explanation method to one model for one period, so the explanation cannot be cross-checked and it is unknown whether the drivers of demand change from year to year.
- **Explanations that stay in notebooks.** Explanation outputs are often left as plots in a notebook, disconnected from the dashboards and anomaly alerts that planners actually use.

The problem addressed by this thesis is therefore: *how can the predictions of an accurate electricity demand forecasting model be made transparent and trustworthy,*

by combining global and local explanation techniques, analysing how feature influence changes over time, and connecting the explanations to anomaly detection in an interactive tool?

This problem includes several practical challenges: selecting and tuning a model that is suitable for exact explanation; preparing 15-minute time-series data without letting information from the future leak into training; explaining which features drive demand across individual years; explaining individual predictions across low, medium, high and peak demand levels; and delivering the explanations to users and linking them to detected anomalous days.

1.4 Research Gap

Although machine learning has achieved strong forecasting accuracy, the following gaps remain in the context of explainable electricity demand forecasting.

- **Black-box predictions without reasons.** Forecasting models report a demand value but do not tell planners why, which reduces trust and prevents action on unusual predictions [4], [5].
- **Use of a single explanation method.** Many studies rely on either SHAP or LIME alone, so explanations cannot be checked against a second, independent method [1], [2], [7].
- **No analysis of temporal drift.** Explanations are normally produced for one fixed period, so it is unknown whether the influence of weather or economic features changes across years.
- **Weak link between anomaly detection and explanation.** Systems that flag abnormal demand days rarely explain how the forecasting model behaved on those days [9].
- **Limited decision-support delivery.** Explanation outputs are often left as notebook plots rather than being delivered through an interactive dashboard that non-technical staff can use [8].

1.5 Aim and Objectives

The objective is to put in place an Explainable AI module for an electricity demand forecasting model. By rendering the model's predictions intelligible, this component will allow energy planners to see what is driving any given forecast and, in turn, have greater confidence in the results**.**

The aim is pursued through the following objectives:

1. First, the electricity demand dataset is put in order with a chronological train-test split and the removal of outliers via a target-only approach.
2. A comparison is made between Linear Regression, Random Forest and XGBoost on this data; the model that offers the best predictive performance and is most amenable to exact explanation methods will be chosen.
3. The XGBoost model is then optimised with Optuna. Care is taken to leave the test set alone to preclude any form of leakage.
4. For the period 2020 to 2025, SHAP is applied on a year-by-year basis to determine global feature importance and see how contributions from features evolve.
5. LIME is used to make sense of individual predictions for 15 instances that cover a range of demand levels.
6. These explanations from both SHAP and LIME are provided via a FastAPI backend and two interactive dashboards, where they are also tied into the system's anomaly detection.
7. Finally, we put the model through its paces using RMSE, MAE and R^2 to gauge its merit, and evaluate the generated explanations for their reliability and utility.

1.6 Research Questions and Hypotheses

The study is guided by the research questions (RQ) and hypotheses (H) in Table 1.2.

Table 1.2: Research questions and associated hypotheses

ID	Research question	Hypothesis
RQ1	Which of Linear Regression, Random Forest and XGBoost gives the lowest error on the same chronological test set, and can the selected model be tuned without touching the test set?	H1: A tuned XGBoost model predicts the test set with low error and similar training and test error (no overfitting).
RQ2	Which input features drive electricity demand, and does their influence change between 2020 and 2025?	H2: Temperature is the most influential feature in every year, and feature influence shows measurable year-to-year drift.
RQ3	Can LIME give useful local explanations for individual predictions across low, medium, high and peak demand?	H3: LIME identifies the main contributing features for each selected instance, and its local R^2 shows how well the local model matches the XGBoost model.
RQ4	Can the explanations be delivered through an API and dashboard and connected to anomaly detection?	H4: An anomaly flagged on the behaviour dashboard can open the matching LIME explanation page for the same date.

1.7 Scope, Limitations and Assumptions

In scope: The work encompasses a number of tasks, from the preparation of the 15-minute demand dataset to a comparison of three regression models. XGBoost was put through its paces with Optuna tuning. On the analytical side, there is a yearly SHAP analysis covering the period from 2020 to 2025 and LIME explanations for 15 chosen instances. The project also calls for a FastAPI backend with two endpoints, two dashboard pages and a date link to tie in with the behaviour intelligence component.

Out of scope: The research is aimed at enhancing forecasting accuracy for its own sake. In addition to the formal user study and a causal analysis of what drives demand, the work covers the deep-learning models put together by the forecasting component. LIME is applied across the full dataset on a daily basis, with the end result being deployment on an operational utility system.

Assumptions: One can consider the dataset to be a fair representation of the demand the system is called on to handle in the course of operation, and the data's chronological sequence is sound. As for the SHAP and LIME values, they are

indicative of the model's behaviour rather than the actual drivers of electricity demand in the real world. Section 7.5 provides a discussion of the principal limitations.

1.8 Individual Contribution

The individual contributions of the author to the project are summarised in Table 1.3.

Table 1.3: Individual contributions of the author

Area	Contribution
Data preparation	Datetime conversion and sorting, year extraction, target-only IQR outlier removal and chronological 80/20 split
Model development	Training and comparison of Linear Regression, Random Forest and XGBoost on identical data
Model tuning	Optuna search with 50 TPE trials over eight hyperparameters, using a validation subset and leaving the test set untouched
Global explanation	SHAP TreeExplainer analysis for each year from 2020 to 2025 and comparison of mean absolute SHAP values over time
Local explanation	LIME explanations for 15 instances covering low, medium, high and peak demand, with local R^2 for each
Delivery and integration	FastAPI backend, SHAP Temporal Drift and LIME Anomaly Diagnosis dashboard pages, and a date link from the behaviour intelligence dashboard

1.9 Relationship to the Project Proposal

This thesis follows the research proposal submitted in March 2026 but the implementation changed in some places as the work progressed. Table 1.4 lists the main differences, so that readers who compare the two documents can see what was planned, what was built and why.

Table 1.4: Comparison of the project proposal and the implemented component

Aspect	Proposal	This thesis	Reason or status
Model explained	Pre-trained XGBoost, Random Forest, LSTM and GRU models from the forecasting component, with XGBoost preferred	Linear Regression, Random Forest and XGBoost trained and compared within this component; XGBoost tuned with Optuna and explained	A tunable tree model gives exact SHAP values; the deep-learning models are not explained (future work)

Aspect	Proposal	This thesis	Reason or status
Explanation methods	SHAP primary; LIME, permutation importance and partial dependence plots optional	SHAP and LIME both implemented; permutation importance and partial dependence plots not implemented	LIME was made a core method to give the local view; the other two are future work
Special-case analysis	Peak, low, anomaly and public-event days	LIME on 15 instances covering low, medium, high and peak demand; anomaly days reached by a date link;	Scope kept to what could be verified
Temporal analysis	Not planned	SHAP computed for each year from 2020 to 2025	Added to study whether explanations drift
Backend API	Four endpoints (three POST, one GET), TLS and token authentication, 500 ms response target	Two GET endpoints serving pre-computed SHAP and LIME results	Simplified because results are pre-computed
Technology	Flask or FastAPI; React, HTML or JavaScript	FastAPI;	Choice between the options was made
Validation	Explanation fidelity and stability against permutation importance; user evaluation with planners	Model metrics, year-by-year SHAP comparison and LIME local R^2; no permutation-importance check and no user evaluation	Time; reported as limitations and future work
Research approach	Design Science Research	Design Science Research approach with an experimental evaluation (Section 3.1)	Consistent; the user-evaluation step was not done
Causal wording	Proposal speaks of causal insights and cause-and-effect analysis	SHAP and LIME are treated as describing model behaviour, not causes	These methods cannot prove causation
Integration	Behaviour, forecasting and scenario components	Working link to the behaviour dashboard;	Other links are future work
Data	Historical Sri Lankan electricity and weather data prepared by the data component	15-minute dataset shared by the team (file `load_forecasting_dataset_corrected.csv`);	The dataset shows signs of being generated (Section 7.1)

Source: Project proposal, R26-IT010, and this thesis.

1.10 Thesis Structure

A review of the pertinent literature on electricity demand forecasting, temporal drift and explainable AI is provided in Chapter 2, where the research gap is also put into context. The next chapter sets out the methodology for the study, covering the data, procedure and evaluation measures as well as any ethical issues. In Chapter 4 one will find the design and requirements for the XAI element of the overall solution.

Chapters 5 and 6 deal with the practical side of the work; the former covers implementation and testing while the latter puts forward the results and their validation. These are then taken up for discussion in Chapter 7, along with an assessment of the contributions, limitations and what they imply. Matters of commercialisation and sustainability are the subject of Chapter 8. The thesis is brought to a close in Chapter 9 with some thoughts on future directions. A list of references, appendices and a glossary are included at the end.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

The review is a targeted narrative review intended to position the component, not an exhaustive systematic survey. It was organised around the topics *electricity demand forecasting*, *explainable artificial intelligence*, *SHAP and LIME in energy systems*, *load forecasting with anomaly detection* and *gradient boosting for load forecasting*. Priority was given to studies that apply explanation methods to electricity demand or load, and to the original papers that introduced the methods used in this thesis.

Statements about individual studies are limited to what those studies report in their titles and abstracts. Numerical results from different studies are not directly comparable because they use different datasets, horizons and spatial levels, and they are therefore not compared numerically in this thesis.

2.2 Domain and Theoretical Foundations

2.2.1 Electricity demand forecasting

It is standard practice to classify load forecasts according to their horizon, spanning the very short-term of a few minutes or hours out to years for the long-term. While statistical approaches like ARIMA and multiple linear regression are useful for putting a number on the impact of variables like the time of day or temperature, they are not well suited to the complex, non-linear and more volatile patterns one sees in consumption [3], [6]. For that reason, machine-learning techniques have gained favour; Random Forest [20], support vector machines and gradient boosting deal with non-linearities more effectively and have been shown to be effective for short- and medium-term work [3]. Then there are deep-learning options such as LSTM and GRU which can pick up on temporal dependencies – indeed, when pitted against ARIMA and Prophet, an LSTM network has been found to have the edge in short-term forecasting [6].

The difficulty in making an accurate forecast lies in the fact that a host of influences are at play and often overlap: from daily, weekly and annual rhythms to weather, public holidays, the state of the economy and structural shifts. An energy planner will want to make sense of any prediction that seems out of the ordinary given these factors, and it is this need for understanding that underpins the explanation methods we review here.

2.2.2 Gradient boosting and XGBoost

Gradient boosting trains trees sequentially, where each tree aims to minimize the mistakes made by previous trees. XGBoost [12] is a popular, scalable algorithm that includes regularisation to reduce overfitting and can be configured using hyperparameters like tree depth, learning rate, subsampling, and minimum child weight. On the other hand, a Random Forest [20] is the averaging of independent trees. Both algorithms are tree-based models, which is important for this thesis since SHAP values [15] can be efficiently computed in tree ensembles.

2.2.3 Interpretability and explainable AI

Interpretability is the degree to which a human can understand the cause of a model's decision [16]. Simple models such as linear regression are interpretable by design because their coefficients show directly how much each input changes the output. Complex models are usually not, and a large body of work therefore proposes *post-hoc* explanation methods that are applied after a model is trained. Rudin [21] argues that for high-stakes decisions inherently interpretable models should be preferred; where a more complex model is nevertheless used, post-hoc explanations are the practical way of exposing its reasoning.

Post-hoc methods are commonly divided into **global** explanations, which describe how a model behaves over a whole dataset, and **local** explanations, which describe one individual prediction. A trustworthy explanation system benefits from both, and from more than one method so that explanations can be compared.

2.2.4 SHAP

SHAP unifies several earlier explanation methods under one framework based on Shapley values from cooperative game theory [10], [17]. In the original game-theory setting, a Shapley value is the fair share of a prize that a player should receive given their contribution to every possible team that player could join. SHAP treats each input feature as a player and the prediction as the prize: the SHAP value of a feature is the amount, in the units of the prediction, by which that feature moves the prediction away from the average prediction. For tree-based models, the TreeExplainer algorithm computes these values exactly in polynomial time, instead of approximating them [15]. Averaging the absolute SHAP values of a feature over many rows gives a global importance score for that feature.

2.2.5 LIME

LIME explains one prediction at a time [11]. It generates many slightly altered copies of the instance, asks the original model to predict each copy, weights the copies by how close they are to the original instance, and fits a simple linear model to these weighted predictions. The weights of that simple model are the explanation. Because LIME only needs the model's prediction function, it is model-agnostic. Its explanation is only reliable to the extent that the simple local model matches the real model near the instance, which can be reported as a local R^2, and LIME explanations are known to vary with sampling and with the definition of the neighbourhood [18]. For this reason, this thesis reports the local R^2 next to each LIME explanation.

2.2.6 Temporal drift of explanations

Concept drift is a change over time in the relationship between model inputs and the target, which reduces the accuracy of a fixed model [19]. A related question, less often studied, is whether the *explanation* of a model changes over time, that is, whether features that were important in one year become more or less important in another. Computing global explanations separately for successive periods is a simple way of observing this *temporal drift of feature importance*, and it can give planners an early signal that demand behaviour is changing.

2.3 Related Research

2.3.1 Explainable AI for electricity forecasting

Several studies apply explainability to energy problems. Bolstad et al. [1] apply explainable AI to day-ahead load forecasting. Alilu et al. [2] build an explainable AI model for electricity usage prediction in Nigeria. Sakkas et al. [7] study explainable approaches for forecasting building electricity consumption. Pati and Mistry [8] apply explainable AI to demand-side management in a single-phase, multi-type consumer base, Shyam et al. [9] combine electrical load forecasting with anomaly detection and explainability, and Yan and Ruan [24] combine spatiotemporal decoupling with explainable AI for power load and price forecasting. Across these studies, features such as temperature, historical load, time of day and weather conditions are among those reported to influence predictions.

2.3.2 Forecasting and feature analysis studies

A second group of studies concentrates on forecasting accuracy and on identifying the factors behind a forecast. Ahmad et al. [3] survey load forecasting techniques and their research challenges. Hachiya and Kamei [4] predict power demand and identify contributing factors with sparse modelling and ensemble learning. Joshi et al. [5] analyse feature engineering in deep-learning architectures for dynamic electricity price forecasting, the comparison of ARIMA, Prophet and LSTM in [6] shows how model choice affects short-term load forecasting, and Fu et al. [25] address masked multi-step probabilistic forecasting for short- to mid-term demand. Together these studies show that feature choice and model choice both matter, but most of them do not attach an explanation to individual predictions.

Table 2.1: Summary of selected related studies and their relationship to this work

Ref.	Focus of the study (as stated in its title)	Relationship to this work
[1]	Day-ahead load forecasting using explainable AI	Shows XAI applied to load forecasting; this work adds yearly drift analysis and a dashboard link
[2]	Explainable AI model for electricity usage prediction in Nigeria	Another regional XAI application; this work targets Sri Lankan 15-minute demand data
[3]	Survey of load forecasting techniques and challenges	Background on statistical, machine-learning and deep-learning models
[4]	Power demand prediction and factor identification with sparse modelling and ensemble learning	Factor identification motivates explaining which features drive demand

Ref.	Focus of the study (as stated in its title)	Relationship to this work
[5]	Feature engineering in deep learning for electricity price forecasting	Shows the effect of input features on model performance
[6]	Electrical load forecasting using ARIMA, Prophet and LSTM	Background on model families for short-term forecasting
[7]	Explainable approaches for forecasting building electricity consumption	XAI applied to electricity consumption forecasting
[8]	Explainable AI for demand-side management	XAI used to support operational decisions
[9]	Load forecasting with anomaly detection and explainability	Closest to the anomaly-to-explanation link built in this work
[24]	Power load and price forecasting with spatiotemporal decoupling and explainable AI	XAI applied to load and price forecasting in an industrial setting
[25]	Masked multi-step probabilistic forecasting for short to mid-term demand	Probabilistic forecasting; related to the forecasting component

2.4 Existing Systems, Products and Techniques

Table 2.2 compares the families of explanation techniques that were considered. Inherently interpretable models, global importance measures such as permutation importance, and the two post-hoc methods chosen here each give a different kind of answer.

Table 2.2: Comparison of explanation technique families

Technique	Scope	Strength	Weakness
Linear regression coefficients	Global	Directly readable; no extra method needed	Only valid if the real relationship is close to linear
Tree feature importance / permutation importance	Global	Cheap and simple	Does not explain single predictions or give direction of effect
SHAP (TreeExplainer)	Global and local	Exact for tree models, consistent attribution in the units of the prediction [10], [15]	Tied to tree models for exact values; can be slower on large data

Technique	Scope	Strength	Weakness
LIME	Local	Model-agnostic; easy to read for one instance [11]	Depends on sampling and neighbourhood; needs a fidelity check [18]

Existing dashboards for energy forecasting generally display forecasts and errors, and notebooks that contain SHAP or LIME plots are rarely connected to the alerts that tell a planner something unusual has happened. The component in this thesis is designed to fill that gap, as described next.

2.5 Gap Synthesis and Conceptual Framework

The review shows that explainable AI is already being applied to electricity forecasting, but that most work uses one method, one period and a static output. The conceptual framework of this thesis therefore combines (i) a tree-based model that SHAP can explain exactly, (ii) SHAP computed per year to expose temporal drift, (iii) LIME on carefully chosen instances to explain individual predictions with a fidelity measure, and (iv) delivery through an API and a dashboard that is linked to anomaly detection. Table 2.3 maps each gap to the objective and design element that responds to it.

Table 2.3: Mapping of research gaps to objectives and design elements

Research gap	Objective	Design element
Black-box predictions without reasons	1, 2, 4, 5	XGBoost with SHAP and LIME explanations
Use of a single explanation method	4, 5	SHAP for the global view, LIME for the local view
No analysis of temporal drift	4	One SHAP analysis per year, 2020 to 2025
Weak link between anomaly detection and explanation	6	Date link from the behaviour dashboard to the LIME page
Limited decision-support delivery	6, 7	FastAPI endpoints and two dashboard pages

3 RESEARCH METHODOLOGY

3.1 Research Design

The research follows a Design Science Research (DSR) approach, as stated in the project proposal: an artefact (the XAI component) is designed, built and evaluated, and the evaluation is carried out as an experimental machine-learning study. It develops and evaluates an explainable-AI component for electricity demand forecasting, and it concentrates on making predictions understandable rather than on improving forecasting accuracy alone. The work proceeds in the stages shown in Figure 3.1: data preparation, comparison of candidate models, tuning of the selected model, global explanation with SHAP, local explanation with LIME, and delivery through an API and dashboard that is linked to anomaly detection. The effectiveness of the approach is judged with predictive metrics for the model and with explanation-based analysis (feature rankings, drift between years and local fidelity) for SHAP and LIME.

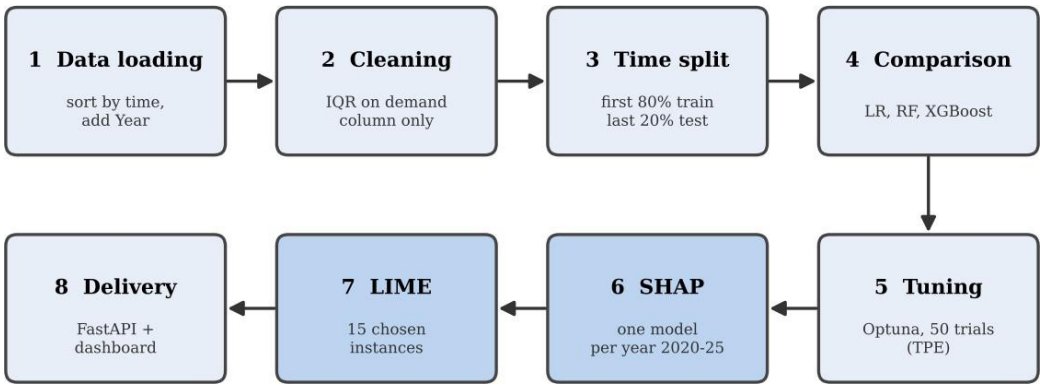


Figure 3.1: Experimental workflow of the XAI component

Source: Author. The two explanation stages (SHAP and LIME) are shaded.

3.2 Data and Study Materials

The study uses the file `load_forecasting_dataset_corrected.csv`, a corrected version of the `load-forecasting-dataset` obtained from Kaggle, which is stored in the data folder of the behaviour intelligence component and shared by all four components of the project.

The file has no missing values and complete 15-minute coverage (35,040 to 35,136 rows in each full year, and 14,496 rows in 2025 up to 31 May). Table 3.1 summarises it.

Table 3.1: Summary of the dataset and study configuration

Item	Description
Records	189,888 observations
Sampling interval	15 minutes
Period	January 2020 to May 2025
Target variable	Electricity demand in kilowatts (kW)
Columns	16 columns after the Year column is added, of which 12 are used as input features (Section 3.2.1)
Input features	12 features: weather, economic and time-related variables (Section 3.2.1)
Rows after outlier removal	188,607 (1,281 rows, 0.67%, removed)
Training and test rows	150,885 training rows and 37,722 test rows
Outlier handling	Interquartile range (IQR) rule, applied to the demand column only
Split	Chronological: first 80% of rows for training, last 20% for testing
Random seed	42 for models, the Optuna sampler and sampling steps

3.2.1 Input features

Every row contains the demand value together with the features in Table 3.2. Weather variables describe the physical conditions that drive heating and cooling load; economic variables describe the wider environment; and time variables allow the model to learn the daily, weekly and annual cycles and the effect of public events.

Table 3.2: Input features used by the model

Group	Features	Role
Weather	Temperature, humidity, wind speed, rainfall, solar irradiance	Physical drivers of cooling, heating and lighting load
Economic	GDP, per capita energy use, electricity price	Slowly changing economic conditions
Time	Hour of day, day of the week, month, public-event indicator	Daily, weekly and annual cycles and holiday effects

3.2.2 Preprocessing, outlier removal and splitting

The timestamp column is read as plain text and is converted to a true date-time value so that it can be sorted and compared correctly. The rows are then sorted into time order, which is essential for the chronological split, and a Year column is created so that the data can later be grouped by year for the SHAP analysis.

Outliers are removed only from the demand column, using the interquartile range rule. If Q1 and Q3 are the 25th and 75th percentiles of demand and IQR = Q3 − Q1, any row with demand below Q1 − 1.5 × IQR or above Q3 + 1.5 × IQR is removed. For this dataset Q1 = 1,365.22 kW, Q3 = 1,635.07 kW and IQR = 269.85 kW, so the accepted range is 960.45 to 2,039.84 kW and 1,281 rows (0.67%) were removed, leaving 188,607 rows. Weather and economic columns are not filtered, because tree-based models such as XGBoost already cope with unusual feature values and because an extreme temperature or rainfall value may be a genuine condition that the explanation should be able to describe. A handful of extremely unrealistic demand readings, in contrast, could teach the model a wrong idea of a normal day.

The cleaned data are split by time, not at random. The first 80% of rows (150,885) form the training set and the most recent 20% (37,722) form the test set (Figure 3.2). A random split of time-series data would let the model train on readings taken minutes before the readings it is tested on, which makes the test score look better than it truly is. A chronological split avoids this form of information leakage [22] and mimics real use, in which a model predicts a future it has not seen.

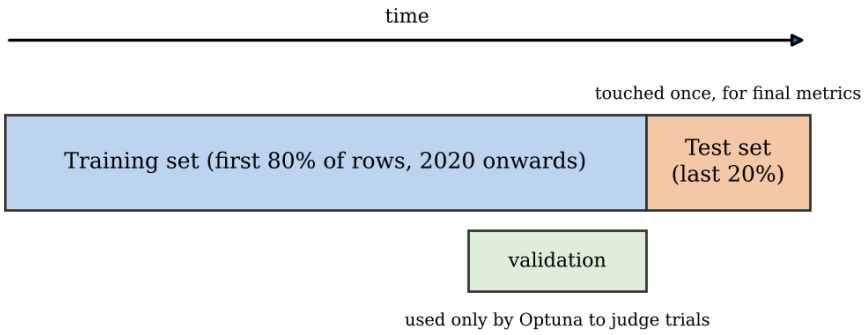


Figure 3.2: Chronological data split and the validation subset used for tuning

Source: Author.

3.3 Experimental Procedure

3.3.1 Model comparison

Three regression models are trained on the same training set and scored on the same untouched test set, so that the comparison is fair: **Linear Regression**, which assumes demand is a straight-line combination of the inputs; **Random Forest** [20], an average of many independently built decision trees; and **XGBoost** [12], a sequence of trees in which each tree corrects the errors of the previous ones. Linear Regression and Random Forest (100 trees, maximum depth 15, random state 42) use the settings given in the code in Section 5.2. Their purpose is to satisfy the need to compare at least two or three models and to show honestly how well the selected model performs against simpler alternatives.

The model that is explained is chosen with two criteria in mind: predictive performance, and compatibility with an exact explanation method. This is described in detail in Section 4.3, because the model chosen for explanation is not necessarily the most accurate one.

3.3.2 Hyperparameter optimisation with Optuna

The selected XGBoost model is tuned with Optuna [13], which tries many combinations of settings automatically and keeps track of which combination performs best. Optuna ran 50 trials using the Tree-structured Parzen Estimator (TPE) sampler [14], a search strategy that uses the results of earlier trials to propose more promising settings, with seed 42. Table 3.3 lists the search space.

Table 3.3: Hyperparameter search space used by Optuna

Hyperparameter	Range	Meaning
n_estimators	100 to 500 (integer)	Number of trees
max_depth	3 to 8 (integer)	Maximum depth of each tree
learning_rate	0.01 to 0.30 (log scale)	Step size with which each tree corrects the previous ones
subsample	0.6 to 1.0	Fraction of training rows used for each tree
colsample_bytree	0.6 to 1.0	Fraction of features used for each tree
min_child_weight	1 to 10 (integer)	Minimum weight needed to create a new tree node
reg_alpha	0.0 to 1.0	L1 regularisation, discourages memorising noise

Hyperparameter	Range	Meaning
reg_lambda	0.0 to 1.0	L2 regularisation, discourages memorising noise

The last 20% of the training rows in time order (30,177 rows) are held back as the validation set, and each trial trains an XGBoost model on a random sample of 10,000 rows taken from the first 80% of the training rows (120,708 rows), to keep the 50 trials fast. Each trial is scored by its RMSE on the validation set. The test set is never shown to Optuna and is used only once, at the end, to report the final performance. The best settings are then used to train the final model on the full training set of 150,885 rows.

3.3.3 Global explanation with SHAP

SHAP TreeExplainer [15] is used to compute, for every row in a sample, how many kilowatts each feature added to or subtracted from the average prediction. The analysis is repeated separately for each year from 2020 to 2025. For each year, a random sample of 2,000 rows of that year's data is drawn (random state 42), an XGBoost model with the tuned settings is fitted to the sample, and the mean of the absolute SHAP values of each feature is taken as that feature's importance for the year. Comparing the yearly importance values shows which features matter most and whether their influence rises or falls over time.

3.3.4 Local explanation with LIME

LIME [11] is applied to the tuned XGBoost model for 15 individual readings. The readings are drawn at random, with a fixed seed, from four demand bands defined by the percentiles of demand in the cleaned data: **low** (below the 25th percentile, 1,366.4 kW), **medium** (25th to 75th percentile, up to 1,634.1 kW), **high** (75th to 90th percentile, up to 1,753.7 kW) and **peak** (above the 90th percentile). Three low, four medium, four high and four peak readings are taken, so that the explanations cover a wide range of situations and not just one type of day. The readings are sampled from the whole cleaned dataset and not only from the test period. A background sample of 2,000 randomly chosen rows lets LIME learn the normal range of each feature, and 1,000 artificial variations are generated around each chosen reading. The model predicts a value for each variation, and a simple linear model fitted to these predictions provides the feature contributions. LIME was run with its default settings, which group each continuous feature into quartile bins. The local R^2 of the simple model is stored with each explanation as a fidelity measure.

3.3.5 Linking explanations to anomaly detection

The behaviour intelligence component identifies days with unusual demand. When an abnormal day is selected, the behaviour dashboard sends the date to the LIME Anomaly Diagnosis page. If the date matches one of the 15 explained readings, that explanation is shown. If it does not, the page still opens, shows the requested date in a banner, and falls back to a default selected instance. This fallback is a known limitation (Section 7.5).

3.4 Measures and Evaluation Metrics

Predictive performance is measured with three metrics on the untouched test set.

- **RMSE** (root mean squared error): the square root of the average squared difference between actual and predicted demand, in kW. It gives more weight to large errors.
- **MAE** (mean absolute error): the average absolute difference between actual and predicted demand, in kW.
- **R^2** (coefficient of determination): the share of the variation in demand that the model explains; 1.0 means a perfect fit.

For the explanations, two further quantities are used. The **mean absolute SHAP value** of a feature in a year is the average of |SHAP value| over the sampled rows and measures how strongly that feature moves predictions in that year. The **local R^2** of a LIME explanation measures how well the simple local model reproduces the XGBoost predictions around the explained instance; a higher value means the explanation is more trustworthy for that instance. Training and test errors of the final model are compared to check for over-fitting, which is when a model memorises the training data and performs worse on new data.

3.5 Data Analysis

The analysis is carried out at two levels. At the global level, mean absolute SHAP values are computed for each feature and each year from 2020 to 2025 and compared to identify the most influential features and any change in their contributions. At the local level, the contributions of individual features are analysed for the 15 selected instances to understand why the model produced each prediction, and the local R^2 is used to judge how far each explanation can be relied upon. Agreement between the two levels (for example, a feature that is globally important and also dominant in local explanations) increases confidence in the explanations, and disagreement is

reported rather than hidden. The results are visualised through the SHAP Temporal Drift and LIME Anomaly Diagnosis dashboard pages.

3.6 Ethics, Privacy, Security and Responsible AI

The study uses an existing electricity-demand dataset. It involves no human participants and collects no personal data, so no personally identifiable information is processed. The dataset consists of aggregated, de-identified demand, weather and economic values, and no output of the component shows information about an individual consumer. The proposal stated that approval from the SLIIT Ethics Review Committee would be sought and that the work would follow the Personal Data Protection Act, No. 9 of 2022 [23].

The component supports responsible AI by showing the reasons behind predictions instead of presenting them as unexplained outputs, and by reporting the local fidelity of each LIME explanation. It is also stated clearly that explanations describe the behaviour of the model, and should not be read as proof of cause and effect in the real electricity system. Data leakage is minimised through the chronological split and by keeping the test set out of hyperparameter tuning. The backend exposes read-only GET endpoints for pre-computed results, which limits the security surface. Any operational use should add authentication, which is provided at the API gateway level in the overall architecture.

3.7 Reproducibility and Research Artefacts

Reproducibility is supported by a documented workflow, fixed random seeds (42), a stated Optuna configuration (50 trials, TPE sampler, seed 42), and the use of the same split for all models. Table 3.4 summarises the reproducibility checklist.

Table 3.4: Reproducibility checklist and status

Item	Status	Note
Dataset described, with size and period	Done	Section 3.2
Dataset source and licence documented	[Confirm]	Add Kaggle link and licence
Chronological split described	Done	Section 3.2.2
Random seeds fixed	Done	Seed 42 for models, sampler and sampling
Item	**Status**	**Note**

Hyperparameter search space and trials stated	Done	Table 3.3, 50 trials
Best hyperparameters recorded	Done	Table A.1
Notebooks and backend code archived	Done (files)	Names in Table A.3;
Library versions recorded	[Confirm]	Add versions of Python, XGBoost, SHAP, LIME, Optuna, FastAPI

The main research artefacts are the two notebooks (the first for data preparation, model comparison, tuning and SHAP; the second for LIME), the saved explanation results, the FastAPI backend and the dashboard pages.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

4.1.1 Stakeholders

The main stakeholders are energy planners and analysts at utilities such as the Ceylon Electricity Board (CEB) and Lanka Electricity Company (LECO), who need to understand and justify demand predictions; operators who look at flagged anomalous days; the other three members of the project team, whose components consume or supply data to this one; and the project supervisors, who assess the research.

4.1.2 Functional requirements

Table 4.1: Functional requirements of the XAI component

ID	Requirement	Objective
FR1	Prepare the demand dataset with target-only outlier removal and a chronological split	1
FR2	Train and compare Linear Regression, Random Forest and XGBoost on identical data	2
FR3	Tune XGBoost without using the test set	3
FR4	Compute SHAP feature importance for each year from 2020 to 2025	4
FR5	Compute LIME explanations, with local R^2, for 15 instances covering low, medium, high and peak demand	5
FR6	Expose SHAP and LIME results through API endpoints	6
FR7	Display year-wise feature influence and per-instance explanations on dashboard pages	6
FR8	Open the matching LIME explanation when an anomaly date is selected on the behaviour dashboard	6

4.1.3 Non-functional requirements

Table 4.2: Non-functional requirements of the XAI component

Quality	Requirement
Usability	Explanations are presented in a format a non-technical planner can read, with feature names and contributions in kW
Reliability	The same inputs and seeds reproduce the same results; a date that has no exact LIME match does not break the page

Quality	Requirement
Maintainability	Machine-learning processing, backend and user interface are kept separate so each can change independently
Performance	Dashboard pages load pre-computed results rather than recomputing explanations on request
Transparency	The page for each explanation shows its source method (SHAP or LIME) and the local fidelity for LIME

4.2 Component Architecture within the Complete Solution

The XAI component is one layer of the larger system in Figure 1.1. It does not replace the forecasting or the anomaly detection; it adds an explanation layer on top of the demand model. Figure 4.1 shows its internal structure. The first notebook prepares the data, compares and tunes the models and computes the yearly SHAP values. The second notebook computes the LIME explanations after the model has been trained and saved. Both save their results, and the FastAPI backend serves them through two GET endpoints, one for SHAP temporal drift and one for LIME anomaly diagnosis, to the corresponding dashboard pages. The behaviour intelligence dashboard sends an anomaly date to the LIME page.

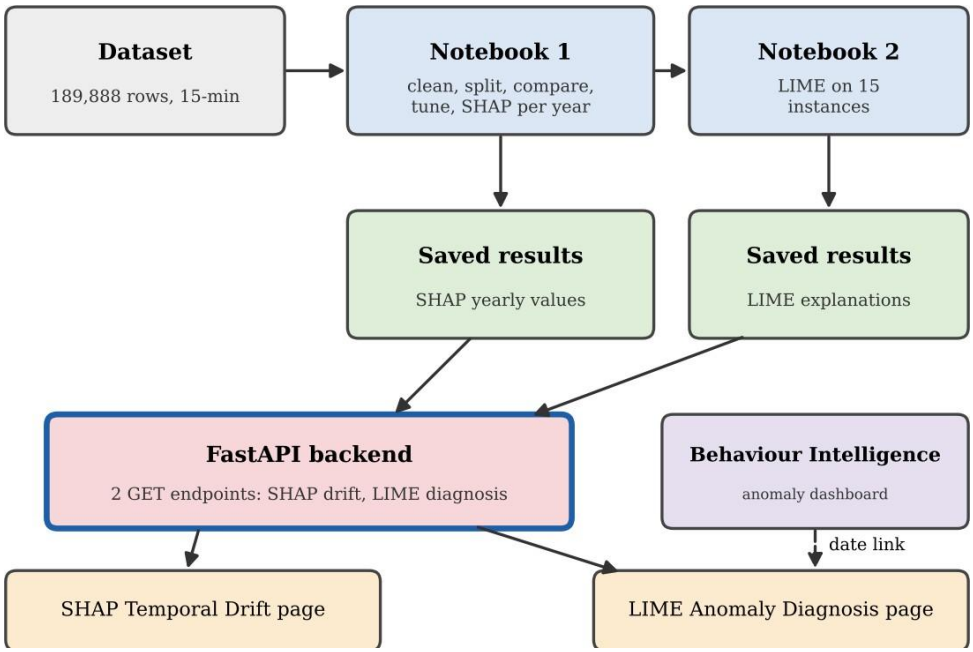


Figure 4.1: Run-time architecture of the XAI component

Source: Author.

4.3 Design Alternatives and Decisions

Table 4.3 records the key design decisions and the reasons for them.

Table 4.3: Key design decisions and justification

Decision	Alternative considered	Justification
Explain an XGBoost model	Explain Linear Regression, which scored best	Linear Regression is already readable from its coefficients, so explaining it adds no research contribution; XGBoost is tree-based, so SHAP TreeExplainer gives exact values
Use SHAP and LIME together	Use only one method	SHAP gives the global view and LIME gives the local view; two methods can be compared
Compute SHAP separately per year	One SHAP analysis for all years	Makes temporal drift of feature importance visible
Chronological split	Random split	Prevents leakage in time-series data and gives an honest test score
Remove outliers from demand only	Remove outliers from all columns	Tree models handle unusual feature values, which may be real conditions; only unrealistic demand values distort learning
Tune with Optuna on a validation subset	Manual tuning or tuning on the test set	Automatic, repeatable search that keeps the test set untouched
LIME on 15 chosen instances	LIME on every row or on demand	Keeps computation small and covers all demand levels; on-demand LIME is future work
Pre-computed results served by FastAPI	Computing explanations at request time	Fast pages, simple deployment, separation of processing from interface

4.4 Detailed Design

4.4.1 Data and model module

This module loads the dataset, converts timestamps, sorts the rows, creates the Year column, removes demand outliers with the IQR rule, performs the 80/20 chronological split and trains the three comparison models. It then runs the Optuna study and trains the final XGBoost model on the full training set. The module writes the evaluation metrics used in Chapter 6.

4.4.2 SHAP temporal module

For each year from 2020 to 2025 the module draws a sample of that year's rows, fits an XGBoost model with the tuned settings, creates a TreeExplainer and computes SHAP values. The mean absolute SHAP value of every feature is stored per year. The output is a table of features by years that the SHAP Temporal Drift page displays.

4.4.3 LIME module

The module builds a LIME tabular explainer in regression mode from a 2,000-row background sample. For each of the 15 chosen instances, it generates about 1,000 perturbed samples, asks the trained XGBoost model for predictions, fits the local linear model and stores the feature contributions, the predicted value, the actual value and the local R^2.

4.4.4 Service interface

The backend is built with FastAPI and provides two GET endpoints: one returns the SHAP temporal drift data and one returns the LIME anomaly diagnosis data.

4.4.5 Dashboard pages

The **SHAP Temporal Drift** page shows the importance of each feature for each year so that users can see which features matter most and how that changes from 2020 to 2025. The **LIME Anomaly Diagnosis** page shows, for a selected instance, the actual and predicted demand, the contribution of each feature and the local R^2. The page accepts a date from the behaviour dashboard, shows it in a banner and selects the matching instance.

4.4.6 Anomaly link

When the behaviour intelligence dashboard flags a day with unusual demand, it shows a button for that day. Selecting it opens the LIME page for that exact date. If the date is one of the 15 explained instances, the explanation is shown directly. Otherwise, the requested date is still displayed and the default instance is shown instead, which keeps the interface working and makes clear that the explanation is not an exact match.

4.5 Standards and Professional Practice

The implementation follows common software-engineering and machine-learning practice: modular components, separation of training and testing data, fixed random

seeds, a documented tuning protocol, and a REST/JSON interface between backend and dashboard. The thesis follows the SLIIT dissertation guidelines for structure and formatting and uses IEEE referencing. Responsible-AI practice is followed by showing explanation fidelity next to each LIME explanation and by stating the limits of what SHAP and LIME can claim.

5 IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

Table 5.1 lists the technologies used. Python is the main language for data processing, model training and explanation. XGBoost provides the forecasting model, Optuna the hyperparameter search, SHAP with TreeExplainer the global explanations and LIME the local explanations. FastAPI serves the results, and a web dashboard presents them.

Table 5.1: Technology stack and justification

Technology	Role and justification
Python	Main language; the machine-learning and explanation libraries are Python libraries
pandas and NumPy	Loading, sorting, cleaning and aggregating the 15-minute data
scikit-learn	Linear Regression, Random Forest and the RMSE, MAE and R^2 metric functions
XGBoost	Gradient-boosted tree model to be explained
Optuna	Automatic hyperparameter search with the TPE sampler
SHAP	TreeExplainer for exact global feature contributions
LIME	LimeTabularExplainer for local explanations
FastAPI	Backend serving the SHAP and LIME results as JSON
Web dashboard [state framework, e.g. React.js]	SHAP Temporal Drift and LIME Anomaly Diagnosis pages

[Add library versions.]

5.2 Component Implementation

Implementation proceeded in the stages below. The most important code blocks are shown to illustrate the logic.

5.2.1 Loading and cleaning the data

```
df = pd.read_csv(DATASET_PATH)
df['Timestamp'] = pd.to_datetime(df['Timestamp'])
df = df.sort_values('Timestamp').reset_index(drop=True)
df['Year'] = df['Timestamp'].dt.year
```

The file is read into a DataFrame, the timestamp is converted into a real date-time value and the rows are sorted into time order. The Year column allows grouping by year for the SHAP analysis.

5.2.2 Target-only outlier removal and chronological split

```
Q1 = df[target_col].quantile(0.25)
Q3 = df[target_col].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

df_clean = df[(df[target_col] >= lower_bound) &
              (df[target_col] <= upper_bound)].copy().reset_index(drop=True)

split_idx = int(len(df_clean) * 0.80)
train_df = df_clean.iloc[:split_idx].copy()
test_df  = df_clean.iloc[split_idx:].copy()
```

Q1 and Q3 mark the boundaries of the middle 50% of demand readings, and rows outside the 1.5 × IQR bounds are removed. Because the rows were already sorted by time, taking the first 80% and the last 20% automatically creates a correct chronological split.

5.2.3 Model comparison

```
lin_model = LinearRegression()
lin_model.fit(X_train, y_train)
lin_pred = lin_model.predict(X_test)

rf_model = RandomForestRegressor(n_estimators=100, max_depth=15,
                                 n_jobs=-1, random_state=42)
rf_model.fit(X_train, y_train)
rf_pred = rf_model.predict(X_test)
```

Linear Regression and Random Forest are trained on exactly the same X_train and y_train as XGBoost and tested on the same X_test, which makes the three results directly comparable.

5.2.4 Tuning with Optuna

```
def objective(trial):
    params = {
        'n_estimators'     : trial.suggest_int('n_estimators', 100, 500),
        'max_depth'        : trial.suggest_int('max_depth', 3, 8),
        'learning_rate'    : trial.suggest_float('learning_rate', 0.01, 0.30,
log=True),
        'subsample'        : trial.suggest_float('subsample', 0.6, 1.0),
        'colsample_bytree': trial.suggest_float('colsample_bytree', 0.6, 1.0),
        'min_child_weight': trial.suggest_int('min_child_weight', 1, 10),
        'reg_alpha'        : trial.suggest_float('reg_alpha', 0.0, 1.0),
        'reg_lambda'       : trial.suggest_float('reg_lambda', 0.0, 1.0),
        'random_state'     : 42,
    }
    model = xgb.XGBRegressor(**params)
    model.fit(X_tr_opt, y_tr_opt)
    preds = model.predict(X_val)
    return float(np.sqrt(mean_squared_error(y_val, preds)))
```

```
study = optuna.create_study(direction='minimize',
        sampler=optuna.samplers.TPESampler(seed=42))
study.optimize(objective, n_trials=50)
```

The objective function is called once per trial. Optuna proposes settings within the given ranges, trains a small XGBoost model, and returns the RMSE on a validation set that is separate from both the tuning training data and the final test set. Optuna keeps the settings with the lowest error.

5.2.5 Final model and evaluation

```
final_model = xgb.XGBRegressor(**best_params)
final_model.fit(X_train, y_train)

y_pred_test = final_model.predict(X_test)
test_rmse = np.sqrt(mean_squared_error(y_test, y_pred_test))
test_mae  = mean_absolute_error(y_test, y_pred_test)
test_r2   = r2_score(y_test, y_pred_test)
```

5.2.6 SHAP explanation, one analysis per year

```
for year in years:
    year_df = df_clean[df_clean['Year'] == year][feature_cols +
[target_col]].dropna()
    sample  = year_df.sample(min(SAMPLE_SIZE, len(year_df)),
random_state=RANDOM_STATE)

    X_yr = sample[feature_cols].copy()
    X_yr.columns = feature_names
    y_yr = sample[target_col]

    yr_model = xgb.XGBRegressor(**best_params)
    yr_model.fit(X_yr, y_yr)

    explainer   = shap.TreeExplainer(yr_model)
    shap_values = explainer.shap_values(X_yr)
    mean_abs    = np.abs(shap_values).mean(axis=0)
```

For each year the loop draws a sample, fits an XGBoost model with the tuned settings, builds a TreeExplainer and computes one SHAP value per feature per row. The average of the absolute values gives one importance score per feature per year. Doing this separately for every year makes it possible to see whether a feature such as temperature became more or less important over time.

5.2.7 LIME explanation, one analysis per instance

```
explainer = lime.lime_tabular.LimeTabularExplainer(
    training_data = X_background.values,
    feature_names = feature_names,
    mode          = 'regression',
    random_state  = 42
)
```

```
exp = explainer.explain_instance(
    data_row     = instance_vals,
    predict_fn   = predict_fn,
    num_features = len(feature_names),
    num_samples = 1000
)
```

The explainer is built from a 2,000-row background sample so that LIME understands the normal range of each feature. For one chosen instance it creates 1,000 artificial variations, asks the XGBoost model to predict each, and fits a simple linear model whose weights are shown on the dashboard.

5.3 Integration with the Other Components

The XAI component is integrated with the behaviour intelligence component. The behaviour dashboard identifies unusual demand days and, for each, shows a button. Selecting it opens the LIME Anomaly Diagnosis page of this component for the same date. The page shows a banner with the requested date; if it matches one of the 15 explained instances the explanation is shown directly, and otherwise the default instance is shown. The connection is therefore a working link between the two components and not only a design intention.

5.4 Verification and Software Testing

Verification covered the data processing, the model, the explanations, the API and the dashboard. Table 5.2 lists the checks and the evidence available for each.

Table 5.2: Verification checks and available evidence

Check	What was verified	Evidence
Data preparation	Timestamps converted, rows in time order, outliers removed from demand only, 80/20 split	Notebook output
Model evaluation	RMSE, MAE and R^2 for the three models on the same test set	Table 6.1
Check	**What was verified**	**Evidence**
No tuning leakage	Optuna used only a validation subset; test set used once	Section 3.3.2, Table 6.2

SHAP	Yearly feature importance generated for 2020 to 2025	Table 6.3
LIME	Contributions and local R^2 generated for the 15 instances	Table 6.4
API	Two GET endpoints return the SHAP and LIME data	[Add test evidence, e.g. screenshots or request logs]
Dashboard	Pages display the returned data correctly	Figure 5.1
Integration	Anomaly selection opens the LIME page for the same date	[Add evidence]

5.5 Deployment and Operation

In operation, the notebooks are run offline to generate the SHAP and LIME results, which are saved. The FastAPI backend loads these results and serves them to the dashboard, which presents them on the two pages. The SHAP Temporal Drift page lets users compare feature contributions across 2020 to 2025. The LIME Anomaly Diagnosis page lets users examine local feature contributions for selected demand cases. Together with the behaviour intelligence component, the complete system supports both the identification of an abnormal day and an explanation of the associated model behaviour.

6 RESULTS AND VALIDATION

6.1 Research Results

6.1.1 Experimental setting

All results come from the 15-minute dataset of 189,888 observations (January 2020 to May 2025). After target-only IQR outlier removal, 1,281 rows (0.67%) were removed and 188,607 remained. The first 80% of rows in time order (150,885) were used for training and the last 20% (37,722) for testing. All three models were trained on the same training set and scored on the same test set. Hyperparameter tuning used a validation set taken from the end of the training data only, and the test set was used once, at the end.

6.1.2 Model comparison

Table 6.1 compares the three models on the untouched test set. RMSE and MAE measure the typical size of the prediction error in kW (smaller is better), and R^2 measures how much of the variation in demand the model explains (1.0 is a perfect fit).

Table 6.1: Test performance of the three models on the same test set

Model	Test RMSE (kW)	Test MAE (kW)	Test R^2
Linear Regression	< 0.0001	< 0.0001	1.0000
Random Forest	0.0117	0.0056	0.99999996
XGBoost (tuned, selected for explanation)	2.0248	1.0512	0.9999

Source: Author's experiments (notebook 1, Step 6b).

The table must be read carefully. Linear Regression, the simplest model, scored best, and Random Forest was also better than XGBoost. All three models explain almost all of the variation in demand (R^2 of 0.9999 or higher). The result tells us something real about this dataset: demand in it is almost an exact linear function of the inputs, so there was no complicated, non-linear structure left for a more powerful model such as XGBoost to exploit. Section 7.1 discusses what this means.

XGBoost was selected for the explanation work because of what happens after prediction, not because it predicts best. SHAP's TreeExplainer gives exact values for tree-based models, and Linear Regression does not need SHAP at all, since its coefficients already show the effect of each feature.

6.1.3 Tuned model performance

The Optuna study ran 50 trials with the TPE sampler and reached a best validation RMSE of 2.3066 kW. The best settings were 467 trees, a maximum depth of 3, a learning rate of 0.0231, a subsample ratio of 0.9248, a column sample ratio of 0.8936, a minimum child weight of 1, and regularisation values of 0.7887 (L1) and 0.4665 (L2) (Table A.1). These settings were used to train the final XGBoost model on the full training set, which was then scored on the test set once. Table 6.2 and Figure 6.1 show the result.

Table 6.2: Performance of the final tuned XGBoost model

Metric	Training set	Test set
RMSE	2.0437 kW	2.0248 kW
MAE	1.0557 kW	1.0512 kW
R^2	0.9999	0.9999

Source: Author's experiments (notebook 1, Step 6).

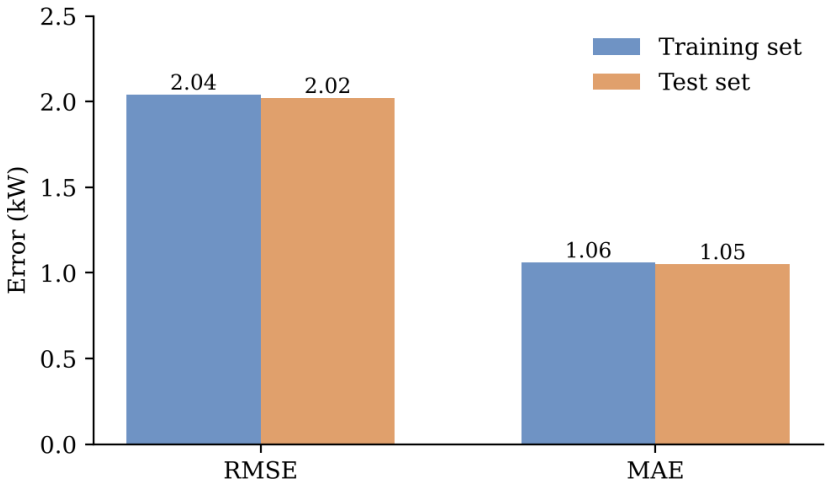


Figure 6.1: Training and test error of the final XGBoost model

Source: Author, from the values in Table 6.2 (rounded to two decimals).

The training and test errors are almost identical (a gap of 0.0189 kW in RMSE), so the model is not over-fitting: if it had memorised the training data, the training error would be much lower than the test error. The test RMSE is 0.14% of the mean test demand of 1,499.65 kW. This supports H1, and it also meets the target set in the proposal that the model being explained should account for at least 90% of the variance in demand.

6.1.4 SHAP results: feature influence and temporal drift

The yearly SHAP analysis shows that **temperature has by far the strongest influence on predicted demand in every year from 2020 to 2025**. Its mean absolute SHAP value was 159.90 kW in 2020, 157.87 kW in 2021, 160.42 kW in 2022, 153.58 kW in 2023, 157.12 kW in 2024 and 157.21 kW in 2025, an average of 157.68 kW. All other features had mean absolute SHAP values of only about 0.2 to 0.3 kW in the years where they ranked second or third, which is several hundred times smaller. The feature that ranked second changed from year to year (wind speed in 2020, 2022 and 2023, electricity price in 2021, GDP in 2024 and rainfall in 2025), which suggests that the order of these minor features carries little meaning. Public event is a special case: its mean absolute SHAP value was exactly zero in 2021 to 2024 and 0.094 kW in 2025, so the per-year models only used it in 2020 and 2025.

Between 2020 and 2025 the influence of temperature fell by 2.69 kW, which is the largest decrease among the features, and the influence of rainfall rose by 0.18 kW, which is the largest increase. Table 6.3 lists the 2020 and 2025 values for all 12 features. The change is not a steady trend, because temperature was highest in 2022 and lowest in 2023, and it is small relative to the size of the effect.

Table 6.3: Mean absolute SHAP value (kW) of the main features, by year

Feature	2020	2025	Observation
Temperature	159.90	157.21	Highest in every year; range 153.58 to 160.42 kW
Rainfall	0.028	0.203	Largest increase, +0.18 kW; second in 2025
Wind speed	0.224	0.066	Ranked second in 2020, 2022 and 2023; fell in 2025
Per capita energy use	0.186	0.091	Third in 2020; lowest in 2022 (0.055)
GDP	0.125	0.192	Highest minor feature in 2024 (0.289 kW)
Humidity	0.160	0.112	Between 0.057 and 0.181 kW; no clear trend

Feature	2020	2025	Observation
Solar irradiance	0.102	0.165	Highest in 2023 (0.201 kW); rose to 0.165 in 2025
Electricity price	0.140	0.121	Second in 2021 (0.197 kW)
Hour of day	0.180	0.041	0.190 kW in 2024; only 0.029 in 2021
Day of week	0.049	0.039	Always below 0.06 kW
Month	0.010	0.011	0.191 kW in 2021; 0.011 in 2020 and 2025
Public event	0.004	0.094	Zero in 2021 to 2024; 0.094 kW in 2025

Source: Author's SHAP analysis (notebook 1, Step 8). All six years are in Table A.2.

6.1.5 LIME results: explanations of individual predictions

LIME was run on 15 instances sampled at random from four demand bands (3 low, 4 medium, 4 high and 4 peak; band limits 1,366.4, 1,634.1 and 1,753.7 kW). For each, the explanation lists the contribution of every feature to the predicted value and the local R^2 of the simple local model. Table 6.4 lists the instances. The XGBoost predictions were close to the actual values, with errors of at most 4.38 kW.

Table 6.4: LIME explanations for the 15 selected instances

ID	Timestamp	Level	Actual (kW)	Predicted (kW)	Error (kW)	Local R^2
I1	2022-11-04 11:15	Low	1187.54	1189.35	+1.81	0.612
I2	2022-02-07 17:30	Low	1331.65	1331.20	−0.45	0.559
I3	2021-09-10 00:45	Low	1155.33	1158.84	+3.51	0.594
I4	2024-03-03 13:30	Medium	1531.94	1532.62	+0.68	0.038
I5	2021-02-09 05:45	Medium	1455.77	1456.29	+0.52	0.051
ID	**Timestamp**	**Level**	**Actual (kW)**	**Predicted (kW)**	**Error (kW)**	**Local R^2**
I6	2020-04-21 16:30	Medium	1505.24	1505.46	+0.22	0.056
I7	2021-08-02 21:45	Medium	1514.29	1513.39	−0.90	0.032
I8	2025-04-26 15:15	High	1639.49	1640.34	+0.85	0.586
I9	2022-10-18 05:30	High	1707.05	1705.45	−1.60	0.616
I10	2023-09-04 00:00	High	1680.55	1680.67	+0.12	0.610

I11	2024-01-10 04:00	High	1647.08	1648.11	+1.03	0.606
I12	2023-01-27 08:45	Peak	1886.34	1882.04	−4.30	0.590
I13	2020-09-15 18:45	Peak	1762.86	1762.79	−0.07	0.607
I14	2023-12-28 11:00	Peak	1757.45	1758.67	+1.22	0.595
I15	2021-03-09 09:30	Peak	1929.97	1925.59	−4.38	0.596

Source: Author's LIME analysis (notebook 2). Error = predicted minus actual.

The local R^2 values fall into two groups. For the 11 low, high and peak instances, the local R^2 was between 0.559 and 0.616 (mean about 0.60), which means the simple local model reproduces a little more than half of the variation of the XGBoost model around those instances. For the four **medium-demand** instances, however, the local R^2 was only 0.032 to 0.056 (mean 0.044), so the LIME explanations for these instances should not be relied upon. The mean local R^2 over all 15 instances is 0.45. A likely reason is the default quartile grouping that LIME applies to continuous features: an instance in an outer band, such as very high temperature, is described well by a feature being in an extreme bin, whereas an instance in a middle band is not. This explanation was not tested in the study.

[Insert screenshot: a LIME bar chart for one peak-demand instance (for example instance I12) from the LIME Anomaly Diagnosis page]

Figure 6.3: Example LIME explanation for a peak-demand instance

Temperature dominated the LIME explanation of every one of the 15 instances. For the low, high and peak instances its weight was between 342 and 363 kW in size (351, 342 and 350 kW for the three low instances; 346 to 359 kW for the high ones; 353 to 363 kW for the peak ones). Every other feature contributed 30 kW or less in

these 11 instances, mostly between 1 and 25 kW, so the picture agrees with the SHAP ranking, in which temperature is first by a wide margin. The second feature differed from instance to instance (public event, wind speed, GDP, per capita energy use, month, solar irradiance), which again suggests that the order of the minor features is not meaningful. For the four medium-demand instances the temperature weight was much smaller (−71, +87, −92 and −74 kW) and other features reached 30 to 69 kW (for example public event in I4 and I6), but these four explanations have a local R^2 of only about 0.04 and should not be interpreted. One further point needs care. In the stored file, the intercept minus the sum of the weights reproduces the model prediction (for I1, 1,620.6 − 421 ≈ 1,199 against a prediction of 1,189), which suggests the weights are saved with the opposite sign to their effect on the prediction. Read this way, a temperature of 24.6 to 26.3 °C (low instances) pushed the prediction down, and 29.4 to 32.3 °C (high and peak instances) pushed it up, which is what would be expected. The sign convention should be confirmed in the notebook before submission. These results support H3 only in part: LIME produced an explanation for every instance, but its fidelity was acceptable only for the low, high and peak groups.

6.2 Product and Technical Results

The component delivered two working dashboard pages and a FastAPI backend with two GET endpoints, one returning the SHAP temporal drift data and one returning the LIME anomaly diagnosis data. The SHAP Temporal Drift page presents year-wise feature contributions from 2020 to 2025. The LIME Anomaly Diagnosis page presents the local explanation of a selected instance together with its local R^2. The two notebooks write their results to `temporal_drift.json`, `anomaly_diagnosis.json` and `xgb_model.pkl`, which the backend serves.

The proposal set two quantitative targets for the component. The first was that the model being explained should account for at least 90% of the variance in demand; the tuned XGBoost model reaches a test R^2 of 0.9999, which meets this target. The second was that explanation results should reach the front end within 500 ms; this was not measured.

6.3 User or Stakeholder Validation

No formal user study or stakeholder evaluation was carried out for this component. Evidence on whether planners find the explanations understandable and useful is therefore not available, and any such claim is treated as untested.The proposal planned a user evaluation with energy planners using structured tasks and questionnaires, and this was not completed. A usability evaluation is proposed as future work.

6.4 Integration Results

The component is connected to the behaviour intelligence dashboard. When an anomalous demand day is flagged there, selecting its button opens the LIME Anomaly Diagnosis page for the same date; the page shows the requested date in a banner and displays the matching explanation if the date is one of the 15 explained instances. If the date is not one of them, the page still opens but shows the default selected instance instead of an exact match. This supports H4 and also marks the main current limitation of the integration.

6.5 Summary of Findings

Table 6.5: Summary of findings and strength of evidence

ID	Finding	Strength of evidence
F1	On this dataset, Linear Regression and Random Forest were more accurate than XGBoost, and all three explain almost all variation (R^2 of 0.9999 or higher)	Strong for this dataset; may not hold for noisier real data (Section 7.1)
F2	The tuned XGBoost model has almost the same training and test error (2.0437 and 2.0248 kW RMSE)	Strong; single run on one chronological split
F3	Temperature is by far the most influential feature in every year (153.6 to 160.4 kW); all other features are about 0.2 to 0.3 kW	Strong for the models fitted per year; describes model behaviour, not causation
ID	**Finding**	**Strength of evidence**

F4	Drift between years is small and not steady: temperature falls by 2.69 kW and rainfall rises by 0.18 kW from 2020 to 2025	Weak to moderate; differences are small and the order of minor features changes each year
F5	LIME explained 15 instances; local R^2 is about 0.60 for low, high and peak instances but only about 0.04 for medium instances	Limited to 15 instances; explanations for medium instances are unreliable
F6	An anomaly on the behaviour dashboard opens the LIME page for the same date	Demonstrated; exact match only for the 15 instances
F7	No user or stakeholder validation has been done	Gap

7 DISCUSSION

7.1 Interpretation of Findings

Accuracy is not the point of this comparison. The simplest model scored best, which is unusual for electricity demand, where non-linear effects of temperature, time of day and holidays normally let tree ensembles and neural networks outperform linear models [3], [6]. A test R^2 of 1.0 and an error below 0.0001 kW for Linear Regression means that demand in this dataset is almost an exact linear function of the 12 features. The summary statistics of the dataset suggest why: the features have suspiciously round means and spreads (temperature mean 28.0 °C with standard deviation 2.0, humidity 80.0% with 5.0, wind speed 2.0 m/s with 1.0, solar irradiance 250 W/m² with 50, GDP 1,000 with 50, per capita energy use 500 kWh with 20, electricity price 25 LKR/kWh with 5, and demand 1,500 kW with 200), and some values are physically impossible, such as a minimum wind speed of −2.83 m/s and a maximum humidity of 101.1%. Real measurements rarely look like this, and it is characteristic of data generated from random draws and a formula. The origin of the dataset is not documented in the project,so this thesis treats it as probably synthetic and does not claim that the accuracy figures would carry over to real utility data, where noise, meter faults and unexplained behaviour are normal.

Why explaining XGBoost is still justified. The research question is about explanation, not accuracy. A linear model is already readable from its coefficients, so explaining it would add nothing, while XGBoost gives the explanation methods a model with real internal structure, and SHAP's TreeExplainer gives exact values for it [15]. The decision is stated openly in Section 4.3 so that the model choice is transparent.

Temperature dominates and is stable. SHAP shows that temperature was by far the strongest driver of predicted demand in every year, which agrees with the finding of earlier XAI studies that temperature and time-related features are among the most influential [1], [7], [8]. The yearly values of temperature (153.6 to 160.4 kW) vary without a steady trend, its influence fell by 2.69 kW between 2020 and 2025, and the influence of rainfall rose by 0.18 kW. These are the temporal-drift signal, and they are small: a planner should read them as a gentle shift rather than a structural change, and the order of the minor features (all about 0.2 to 0.3 kW) should not be interpreted

at all. The dominance of one feature over all others is itself a sign of how simple the data are, since in real demand data hour of day, day of week and holidays normally carry visible weight. It is also worth noting that the yearly SHAP values describe an XGBoost model fitted to a sample of each year's data with the tuned settings, so the drift reflects both the data of that year and the refit (Section 7.5).

Two views of one model. SHAP and LIME answer different questions. SHAP gives the big picture, year by year, and LIME gives the close-up view of one reading. Using both lets each be checked against the other, which is part of what makes the explanation trustworthy. The local R^2 of LIME makes the quality of each close-up explicit, and it shows that the fidelity is acceptable (about 0.60) for low, high and peak instances but very poor (about 0.04) for the four medium-demand instances, so those explanations should not be trusted. The most likely cause is LIME's default grouping of continuous features into quartile bins, but this was not tested; switching off the grouping or using more perturbed samples is an obvious next step.

7.2 Comparison with Literature Baselines and Existing Solutions

Earlier studies applied SHAP or LIME to load forecasting [1], electricity usage in Nigeria [2], building consumption [7] and demand-side management [8], and one combined load forecasting, anomaly detection and explainability [9]. This work follows the same direction and extends it in three ways: it compares SHAP across individual years to expose temporal drift, it uses SHAP and LIME together with a fidelity measure for the local explanations, and it connects the explanations to an anomaly dashboard in a working link. Numerical results are not compared with those studies because the datasets, horizons and models differ.

7.3 Research Contribution

- A documented comparison of Linear Regression, Random Forest and XGBoost under one chronological protocol, with an honest account of why the best-scoring model was not the one explained.
- A year-by-year SHAP analysis for 2020 to 2025 that shows the stability of temperature as the dominant driver and small shifts in other features.
- LIME explanations across low, medium, high and peak demand, reported with local fidelity.

- A leakage-aware tuning protocol in which the test set is used only once.

7.4 Product Contribution

The product contribution is an explanation layer that planners can actually open: two dashboard pages backed by a FastAPI service, and a one-click path from a flagged anomaly to its explanation. It turns SHAP and LIME from notebook plots into part of a decision-support system.

7.5 Limitations and Threats to Validity

Table 7.1: Limitations and threats to validity

Limitation	Description	Effect on conclusions
Near-linear, probably synthetic dataset	Linear Regression reaches R^2 of 1.0; feature statistics are suspiciously round; impossible values remain (wind speed −2.83 m/s, humidity 101.1%); provenance undocumented	Accuracy comparison and the dominance of temperature may not transfer to real, noisier data
LIME covers 15 instances only	Explanations exist for 15 randomly sampled readings, not every day; readings come from the whole cleaned dataset, not only from the test period	An anomaly date may fall back to the default instance
Low LIME fidelity for medium demand	Local R^2 is about 0.04 for the four medium-demand instances, against about 0.60 for the others	Explanations for medium-demand instances are unreliable
Tuning on a subsample	Each Optuna trial trained on 10,000 random rows of the first 80% of the training data	Settings may be slightly different from a search on all rows
Yearly SHAP uses per-year models	Each year's SHAP values come from an XGBoost model fitted to a sample of that year	Drift reflects both data change and refitting
Explanation is not causation	SHAP and LIME describe model behaviour; correlated features share credit	Findings should not be read as real-world causes
LIME instability	Explanations depend on random sampling and neighbourhood choice [18]	Mitigated by fixed seed and local R^2, not removed
Single dataset and split	One dataset, one chronological split, single seed	Results may differ on other data
No user evaluation	Usefulness for planners was not tested, although the proposal planned it	Practical benefit is unproven

Limitation	Description	Effect on conclusions
No cross-check with permutation importance	The proposal planned to compare SHAP and LIME with permutation importance; only SHAP and LIME were compared	Stability of the rankings is not independently confirmed
Only the XGBoost model is explained	The deep-learning forecasting models are not covered	Explanations may not describe the models used for forecasting in the system

7.6 Implications

For planners, the results suggest that on this data temperature should be the first thing to check when a prediction looks unusual, and that explanations can be made available without extra effort inside an existing dashboard. For researchers, the near-perfect linear fit is a reminder that benchmark data should be inspected before conclusions about model families are drawn, and that explanation research needs models and data with real complexity. For the project, the explanation layer could be extended to the forecasting component's models, so that the whole system explains the predictions it actually shows.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value Proposition and Target Users

The proposition is an explainable-AI layer instead of a prediction tool alone. It lets planners see *why* a demand prediction is high or low, and why a day was flagged as unusual, so that the AI is treated as a transparent system and not as a black box. The primary customer persona is the **decision analyst**: an energy analyst or planning engineer at a utility such as the Ceylon Electricity Board (CEB). Their pain points are difficulty understanding why demand fluctuates, a lack of transparency in AI predictions, and the inability to justify demand spikes to management. Their goal is to understand demand patterns, identify the main factors behind them and make explainable, data-driven decisions. The explanation service sits beside the forecasting, behaviour and scenario services of the wider system and increases trust in them.

8.2 Market and Competitive Position

The primary market is national energy utilities and grid operators in developing countries, beginning with Sri Lanka and extending to South Asia. Secondary markets are energy regulators, large industrial zones, renewable-energy developers and smart-city planners. The competitive advantage claimed in the proposal is that explainability is a core feature rather than an add-on: the service shows how predictions and diagnoses are made, presents them in a form non-technical users can read, and is designed for local conditions such as regional weather and the effect of holidays. The proposal also described the service as revealing *causal* links between demand drivers; this thesis is more cautious, because SHAP and LIME describe how a model behaves and cannot prove cause and effect (Section 7.5). No formal market study or customer interviews have been carried out, so these claims are hypotheses to be tested in a pilot.

8.3 Business or Sustainability Model

The revenue model in the proposal has three parts: a business-to-government (B2G) annual software-as-a-service subscription for national utilities; business-to-business (B2B) licensing of the explainable-AI module to private energy companies and industrial consumers; and one-off consultancy and integration fees for connecting the

module to existing analytics systems, dashboards or energy-management platforms. Pricing is intended to be value-based and tiered by level of use, number of users and features, with a reduced-price pilot programme so that local utilities can test and confirm the value of the service.

Costs fall into three groups: development costs (explanation methods, data processing and cloud computing), data costs (obtaining, cleaning and updating electricity, weather and event data) and operating costs (server maintenance, model retraining, system updates and technical support). Because the explanations in the implemented component are pre-computed and served as saved results, running costs are modest. Table 8.1 gives the budget planned in the proposal, which indicates the order of magnitude of development cost.

Table 8.1: Planned budget for the XAI component (LKR)

Item	Quantity / duration	Unit cost	Total
Laptop / workstation for model development	1	150,000	150,000
Cloud computing (Google Colab Pro / GPU)	12 months	4,500	54,000
Web hosting / cloud server	12 months	2,500	30,000
Domain and SSL certificate	1	5,000	5,000
Data storage and backup (cloud)	12 months	1,500	18,000
Visualisation and analytics tools	1	6,000	6,000
Data collection / research transport	3	6,000	18,000
Miscellaneous (printing, documentation, internet)	1	10,000	10,000
Total			**291,000**

Source: Project proposal, R26-IT010, Section 7.

8.4 Scalability, Intellectual Property and Compliance

Scalability. Because explanations are pre-computed, the service scales by serving saved results; larger datasets increase only the offline computation, and SHAP's TreeExplainer is efficient for tree models. Extending LIME to on-demand use for

any date would need more computing resources per request. The proposal also envisaged horizontal scaling through containerisation or cloud deployment.

Intellectual property. The proposal lists three forms of protection: the proprietary implementation of the explanation techniques for electricity demand, trade secrets covering the internal logic of explanation generation and visualisation, and software copyright over the source code of the module and its reporting interface. SHAP, LIME, XGBoost and Optuna are open-source tools, so the protectable value lies in the integrated pipeline, the dashboard and the anomaly-to-explanation link, and not in the algorithms. Ownership under SLIIT policy and any data-provider terms should be confirmed.

Compliance. The dataset is aggregated and contains no personal data. A commercial deployment should respect applicable data-protection law, for example Sri Lanka's Personal Data Protection Act, No. 9 of 2022 [23], contractual limits on the use of utility data, and the requirements of the sector regulator. Explanations should always be presented with their limits, so that they are not mistaken for proof of cause.

8.5 Contribution to the Sustainable Development Goals

The proposal aligns the project with SDG 7. This thesis also notes two further goals.

- **SDG 7 (Affordable and Clean Energy).** Understandable, transparent forecasts support efficient and reliable management of the electricity grid and make it easier to plan reserves and integrate variable renewable generation.
- **SDG 9 (Industry, Innovation and Infrastructure).** The component applies explainable AI to critical energy infrastructure.
- **SDG 13 (Climate Action).** Better-trusted forecasts can reduce unnecessary over-generation and so lower emissions. The size of this effect has not been measured in this study.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis presented the Explainable AI component of the AI-Based Electricity Demand Intelligence System for Sri Lanka. Using a 15-minute dataset of 189,888 observations from January 2020 to May 2025, with target-only outlier removal and a chronological 80/20 split, three models were compared. Linear Regression and Random Forest were more accurate than XGBoost on this dataset, which turned out to be close to linear, but XGBoost was selected for explanation because SHAP's TreeExplainer gives exact values for tree models. The tuned XGBoost model, optimised with 50 Optuna trials without touching the test set, reached RMSE 2.02 kW, MAE 1.05 kW and R^2 of 0.9999 on the test set, with training and test errors almost identical. The dataset itself shows signs of being generated, which explains why a linear model fits it perfectly.

SHAP analysis for each year from 2020 to 2025 showed that temperature is the most influential feature in every year, at about 160 kW in 2020 and about 157 kW in 2025, with all other features at only about 0.2 to 0.3 kW and only small, unsteady drift between years. LIME explained 15 instances sampled from low, medium, high and peak demand bands, with a local R^2 of about 0.60 for the low, high and peak instances but only about 0.04 for the four medium-demand instances, whose explanations are therefore unreliable. The explanations are served through a FastAPI backend to two dashboard pages and connected to the behaviour intelligence dashboard, so that a flagged anomalous day opens its explanation. The main limitations are that LIME covers only 15 instances, that the dataset is near-linear and of undocumented origin, and that no user evaluation has been done.

9.2 Achievement of Objectives

Table 9.1: Achievement of the objectives

No.	Objective (abridged)	Status	Evidence
1	Prepare data with target-only outlier removal and chronological split	Achieved	Sections 3.2.2 and 5.2
2	Compare Linear Regression, Random Forest and XGBoost	Achieved	Table 6.1
3	Tune XGBoost with Optuna without test leakage	Achieved	Section 3.3.2, Table 6.2

No.	Objective (abridged)	Status	Evidence
4	Yearly SHAP analysis, 2020 to 2025	Achieved	Table 6.3, Table A.2
5	LIME for 15 instances across demand levels	Achieved, with low fidelity for medium-demand instances	Table 6.4
6	Deliver through FastAPI and dashboard; integrate with anomaly detection	Achieved, with the fallback limitation	Sections 6.2 and 6.4
7	Evaluate model and assess explanations	Partly achieved: model metrics and local R^2 reported; no user evaluation	Sections 6.1 and 6.3

9.3 Individual Research and Product Contributions

The individual research contributions are the leakage-aware model comparison and tuning protocol, the year-by-year SHAP analysis that exposes temporal drift, and LIME explanations with local fidelity across demand levels. The product contribution is the explanation layer of FastAPI endpoints and two dashboard pages and its one-click link from the behaviour intelligence dashboard.

9.4 Future Directions

- **On-demand LIME.** Run LIME for any date chosen by the user instead of only 15 fixed instances, so that every anomaly has an exact explanation.
- **Explain the forecasting models.** Extend the explanation layer to the deep-learning models of the forecasting component, for example with model-agnostic SHAP variants, so that the system explains the predictions it actually displays.
- **More model families.** Compare additional model types such as ARIMA or LSTM, as mentioned in the original project proposal.
- **Real utility data.** Test the pipeline on measured, noisier data from a utility, where non-linear models are expected to matter more.
- **Richer explanations.** Add SHAP interaction values and counterfactual explanations such as *what change in temperature would remove the peak*.

- **More explanation methods and plots.** Add permutation importance and partial dependence plots as planned in the proposal, together with force and waterfall plots and a dedicated public-event day analysis.
- **User evaluation.** Test the dashboard with planners, using structured tasks and questionnaires as planned in the proposal, to measure understanding, trust and usefulness.

REFERENCES

[1] D. A. Bolstad, U. Cali, M. Kuzlu and U. Halden, "Day-ahead load forecasting using explainable artificial intelligence," in *2022 IEEE Power & Energy Society Innovative Smart Grid Technologies Conference (ISGT)*, New Orleans, LA, USA, 2022.

[2] G. E. Alilu, G. C. Okorie, I. A. Ayoade, U. O. Obonna, O. K. Akinde and C. I. Nwakanma, "Explainable artificial intelligence model for electricity usage prediction in Nigeria," in *2024 IEEE 5th International Conference on Electro-Computing Technologies for Humanity (NIGERCON)*, Ado Ekiti, Nigeria, 2024.

[3] N. Ahmad, Y. Ghadi, M. Adnan and M. Ali, "Load forecasting techniques for power system: research challenges and survey," *IEEE Access*, vol. 10, 2022.

[4] T. Hachiya and K. Kamei, "Prediction of power demand and factor identification using sparse modeling AI with ensemble learning for improvement of accuracy," in *2024 Joint International Conference on Soft Computing and Intelligent Systems and International Symposium on Advanced Intelligent Systems (SCIS&ISIS)*, Himeji, Japan, 2024.

[5] P. Joshi, J. Bhardwaj, S. Størdal, M. Kolhe and E. Haugom, "Analysis of feature engineering in deep learning architectures for dynamic electricity price forecasting," in *2025 International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA)*, Antalya, Turkiye, 2025.

[6] D. P. A. et al., "Electrical load forecasting using ARIMA, Prophet and LSTM networks," *International Journal of Electrical and Electronics Research (IJEER)*, 2021.

[7] N. Sakkas et al., "Explainable approaches for forecasting building electricity consumption," *MDPI*, Basel, Switzerland, 2023.

[8] U. Pati and K. D. Mistry, "Explainable artificial intelligence approach for demand-side management in a 1-phase multi-type consumer base: enhancing efficiency and transparency," *IEEE Transactions on Emerging Topics in Computational Intelligence*, vol. 9, no. 4, 2024.

[9] S. S. Shyam et al., "Electrical load forecasting with anomaly detection and explainability," *IJRASET – International Journal for Research in Applied Science & Engineering Technology*, 2025.

[10] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems 30 (NeurIPS)*, 2017.

[11] M. T. Ribeiro, S. Singh and C. Guestrin, "'Why should I trust you?': Explaining the predictions of any classifier," in *Proc. 22nd ACM SIGKDD*

International Conference on Knowledge Discovery and Data Mining, 2016, pp. 1135–1144.

[12] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794.

[13] T. Akiba, S. Sano, T. Yanase, T. Ohta and M. Koyama, "Optuna: A next-generation hyperparameter optimization framework," in *Proc. 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2019, pp. 2623–2631.

[14] J. Bergstra, R. Bardenet, Y. Bengio and B. Kégl, "Algorithms for hyper-parameter optimization," in *Advances in Neural Information Processing Systems 24 (NeurIPS)*, 2011.

[15] S. M. Lundberg et al., "From local explanations to global understanding with explainable AI for trees," *Nature Machine Intelligence*, vol. 2, no. 1, pp. 56–67, 2020.

[16] F. Doshi-Velez and B. Kim, "Towards a rigorous science of interpretable machine learning," arXiv:1702.08608, 2017.

[17] L. S. Shapley, "A value for n-person games," in *Contributions to the Theory of Games, Volume II*, Princeton University Press, 1953, pp. 307–317.

[18] D. Alvarez-Melis and T. S. Jaakkola, "On the robustness of interpretability methods," arXiv:1806.08049, 2018.

[19] J. Gama, I. Žliobaitė, A. Bifet, M. Pechenizkiy and A. Bouchachia, "A survey on concept drift adaptation," *ACM Computing Surveys*, vol. 46, no. 4, 2014.

[20] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[21] C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," *Nature Machine Intelligence*, vol. 1, pp. 206–215, 2019.

[22] S. Kaufman, S. Rosset, C. Perlich and O. Stitelman, "Leakage in data mining: Formulation, detection, and avoidance," *ACM Transactions on Knowledge Discovery from Data*, vol. 6, no. 4, 2012.

[23] Parliament of the Democratic Socialist Republic of Sri Lanka, *Personal Data Protection Act, No. 9 of 2022*, 2022.

[24] C. Yan and H. Ruan, "Power load and price forecasting based on spatiotemporal decoupling and explainable artificial intelligence interaction perception," in *2025 5th International Conference on Electrical Engineering and Mechatronics Technology (ICEEMT)*, Shenzhen, China, 2025.

[25] Y. Fu et al., "Masked multi-step probabilistic forecasting for short to mid term electricity demand," arXiv, 2023.

GLOSSARY

Anomaly – A day or reading whose electricity demand is unusual compared with normal behaviour.

Background sample – The set of rows (here 2,000) from which LIME learns the normal range of each feature.

Chronological split – Division of a time series in time order, so that the model is never trained on data later than the data it is tested on.

Data leakage – Use of information from outside the training period when building a model, which makes results look better than they are.

Global explanation – An explanation of how a model behaves over a whole dataset.

Hyperparameter – A model setting chosen before training, such as the number of trees or the learning rate.

IQR (interquartile range) – The distance between the 25th and 75th percentiles; used to define outlier boundaries.

Local explanation – An explanation of one individual prediction.

Local R^2 – How well LIME's simple local model reproduces the real model's predictions around the explained instance.

Optuna – A tool that automatically searches for the best hyperparameters.

Over-fitting – When a model memorises the training data and performs worse on new data.

SHAP value – The amount, in kW, by which a feature moves a prediction away from the average prediction.

Temporal drift – Change over time, here in the influence of features on predicted demand.

TPE (Tree-structured Parzen Estimator) – A search strategy that uses earlier trial results to propose better hyperparameters.

TreeExplainer – The SHAP algorithm that computes exact SHAP values for tree-based models.

APPENDICES

Appendix A Extended Results and Evidence Record

A.1 Dataset requirements stated in the project proposal

The proposal (Section 3.1) listed the inputs the explanation module would use. Table A.0 checks them against the dataset that was used.

Table A.0: Input requirements in the proposal and the dataset used

Input stated in the proposal	Present?	Note
Temporal information: hour, day, month	Yes	Hour of Day, Day of Week, Month
Weather: temperature, humidity, rainfall	Yes	Also wind speed and solar irradiance
Public events	Yes	Public Event indicator
Lagged load values	No	None of the 12 features is a lagged demand value
Economic variables	Added	GDP, per capita energy use and electricity price were in the dataset

A.2 Input features and best hyperparameters

Table A.1: Input features and best Optuna hyperparameters

Item	Value
Features (12)	Temperature (°C), Humidity (%), Wind Speed (m/s), Rainfall (mm), Solar Irradiance (W/m²), GDP (LKR), Per Capita Energy Use (kWh), Electricity Price (LKR/kWh), Day of Week, Hour of Day, Month, Public Event
n_estimators	467
max_depth	3
learning_rate	0.0231
subsample	0.9248
colsample_bytree	0.8936
min_child_weight	1
reg_alpha	0.7887
reg_lambda	0.4665

Item	Value
Best validation RMSE	2.3066 kW (50 trials)

A.3 Mean absolute SHAP values by year

Table A.2: Mean absolute SHAP value (kW) of each feature, 2020 to 2025

Feature	2020	2021	2022	2023	2024	2025
Temperature	159.90	157.87	160.42	153.58	157.12	157.21
Rainfall	0.028	0.063	0.118	0.035	0.137	0.203
Wind speed	0.224	0.118	0.236	0.301	0.126	0.066
Per capita energy use	0.186	0.106	0.055	0.181	0.118	0.091
GDP	0.125	0.121	0.197	0.130	0.289	0.192
Humidity	0.160	0.140	0.070	0.181	0.057	0.112
Solar irradiance	0.102	0.104	0.124	0.201	0.153	0.165
Electricity price	0.140	0.197	0.067	0.083	0.138	0.121
Hour of day	0.180	0.029	0.086	0.027	0.190	0.041
Day of week	0.049	0.046	0.052	0.008	0.046	0.039
Month	0.010	0.191	0.066	0.073	0.070	0.011
Public event	0.004	0.000	0.000	0.000	0.000	0.094

Source: `temporal_drift.json`, produced by notebook 1, Step 8.

A.4 Evidence sources in the project repository

Table A.3: Project artefacts that hold the reported evidence

Artefact	Content	Location
`1_shap_temporal_drift.ipynb`	Data preparation, model comparison, Optuna tuning, yearly SHAP	madushani_xai/notebooks
`2_lime_anomaly_diagnosis.ipynb`	LIME explanations for 15 instances	madushani_xai/notebooks
`temporal_drift.json`, `anomaly_diagnosis.json`, `xgb_model.pkl`	Saved SHAP results, LIME results and the tuned model	madushani_xai/backend/outputs

Artefact	Content	Location
Dashboard pages	SHAP Temporal Drift and LIME Anomaly Diagnosis	[insert]

Appendix B Academic Integrity and AI Use Disclosure

In line with the module policy, this appendix discloses the use of generative AI. The author remains responsible for the accuracy, originality, citations, data and conclusions of the thesis.

Table A.4: Disclosure of generative AI use

Item	Disclosure
Tool	Claude (Anthropic), a generative AI assistant
Purpose	Structuring and drafting thesis text, tables and diagrams from the author's component summary, draft thesis and project results, following the structure of the SLIIT thesis format
Material provided to the tool	The author's draft thesis, a component summary containing the results, code excerpts and explanations, and an example thesis used only for its structure and formatting
Verification performed	[Describe how every statement, number, citation and figure was checked against project records and the original sources]
Sections affected	[State the sections drafted or edited with AI assistance]
Content not generated by AI	Experimental data, results and measurements; these come from the author's own experiments